

(b) $AB'CD' + A'B'CD + ABC'D + A'BC'D$

$\Sigma(10, 6, 9, 5) = F$

$F = \Sigma(0, 1, 2, 3, 4, 7, 8, 11, 12, 13, 14, 15)$

AB \ CD	00	01	11	10
00	1	1	1	1
01	1		1	
11	1	1	1	1
10	1		1	

$F = C'D' + A'B' + AB + CD$

$F = (F')'$ (Thm 3)

$= (C'D' + A'B' + AB + CD)'$

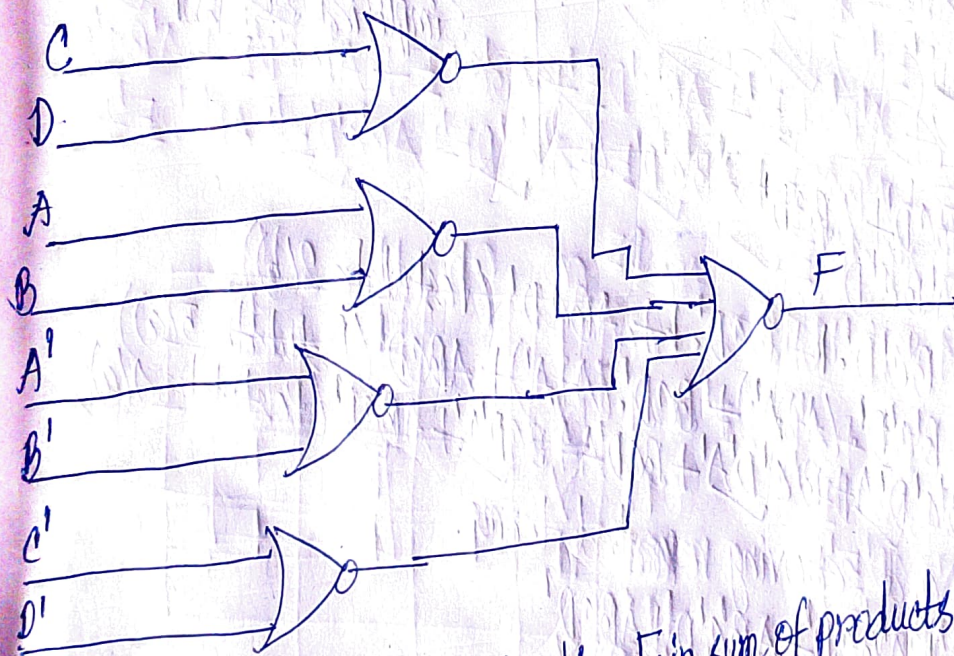
$= (C'D')' \cdot (A'B')' \cdot (AB)' \cdot (CD)'$

(Extended Thm 5(a))

$= (C'' + D'') \cdot (A'' + B'') \cdot (A'B)' \cdot (C'D)'$

(Extended Thm 5(b))

$= (C + D) \cdot (A + B) \cdot (A'B)' \cdot (C'D)'$



3-15: Simplify the Boolean function F in sum of products using the don't-care conditions d :

(a) $F = y' + x'z'$
 $d = yz + xy$

Soln:

$F = y' + x'z' = y' \cdot 1 + x'z'$ (Postulate 2(b))

$= y' \cdot (x + x') + x'z'$ (Postulate 5(a))

$= y'x + y'x' + x'z'$ (Postulate 4(a))

$= y'x + y'x' + x'z' \cdot 1$ (Postulate 2(b))

$= y'x \cdot 1 + y'x' \cdot 1 + x'z' \cdot 1$ (Postulate 2(b))

$= y'x \cdot (z + z') + y'x' \cdot (z + z') + x'z' \cdot (y + y')$ (Postulate 5(a))

$= y'xz + y'xz' + y'x'z + y'x'z' + x'zy + x'zy'$ (Postulate 4(b))